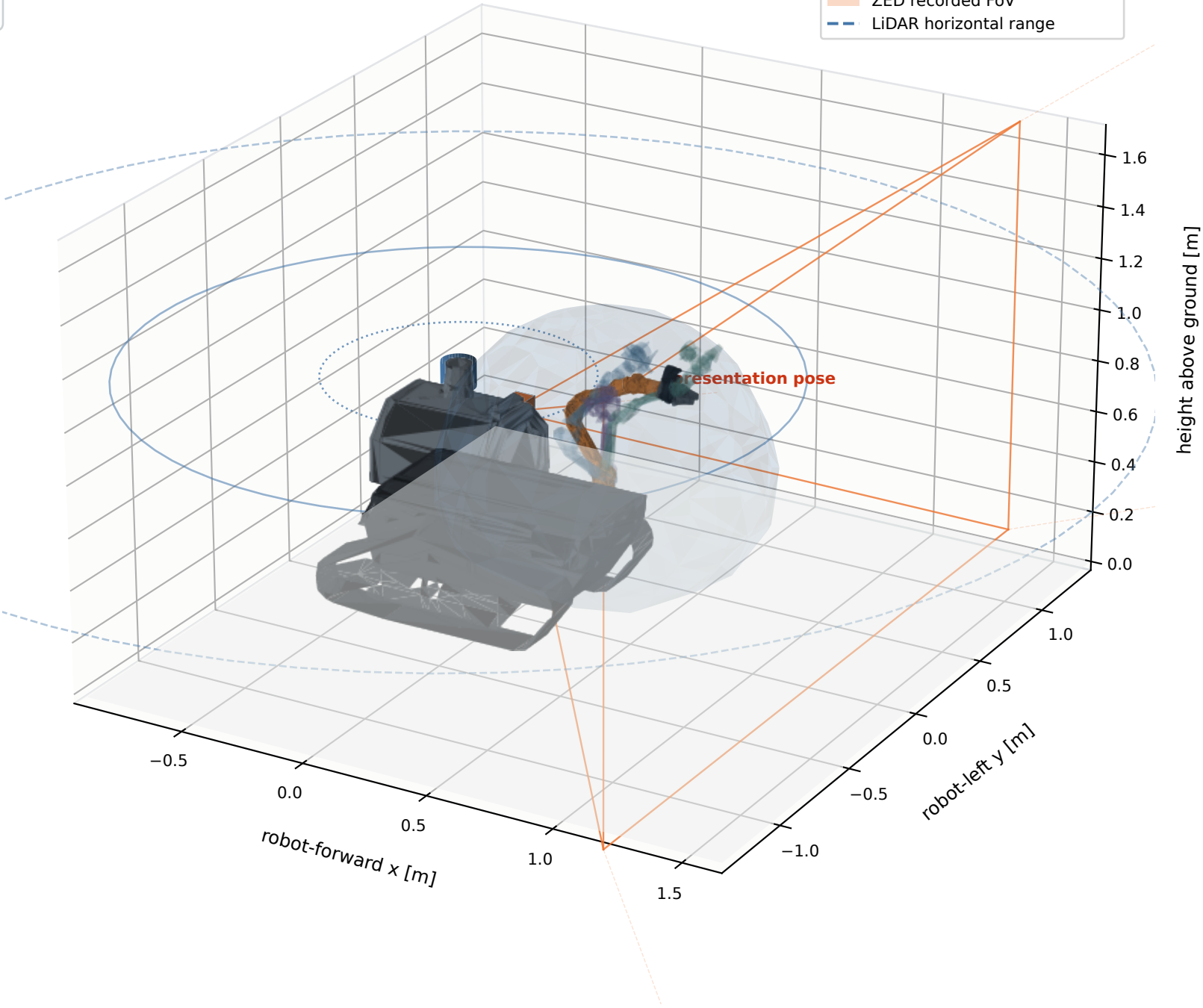


# Integrated 3D Sensing and Manipulation Envelope

SENSING SCALE  
solid near-field: 1.25 m  
dashed continuation: 2.50 m  
configured tracking: 0.5–10.0 m

- Bunker CAD
- PiPER presentation pose
- gripper convex outer envelope
- representative ghost poses
- ZED recorded FoV
- LiDAR horizontal range



| MODEL              | REACH SURFACE                          | ZED INTRINSICS                | LIMIT                                             |
|--------------------|----------------------------------------|-------------------------------|---------------------------------------------------|
| Bunker + PiPER CAD | 30k samples · 0.79 m <sup>3</sup> hull | 1280×720 · H 100.9° · V 68.5° | convex kinematic outer bound · not collision-free |